Winning Streak

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	256 megabytes

Evirir the dragon played N rounds of a video game. The rounds are numbered $1, 2, \ldots, N$ in order of increasing time. Each round, Evirir either won or lost. The results are given as $A_1, A_2, \ldots, A_N$: for each round i, $A_i = 1$ if Evirir won it and $A_i = 0$ if Evirir lost it.

A *winning streak* is a group of consecutive rounds $l, l+1, \ldots, r$ ($1 \le l \le r \le N$) that Evirir all won. Its length is $r - l + 1$.

Evirir wants to brag about its longest winning streak, which is a winning streak with the largest possible length. Formally, Evirir wants to find integers l and r such that:

- $1 \le l \le r \le N$;
- Evirir won round i for all $i = l, l+1, \ldots, r$; and
- $r - l + 1$ is as large as possible.

If there are multiple longest winning streaks, Evirir wants the one with the maximum l. Help Evirir find it!

Input

The first line contains an integer N — the number of rounds.

The second line contains N space-separated integers $A_1, A_2, \ldots, A_N$ — the results of the rounds.

Output

Output two space-separated integers l and r ($1 \le l \le r \le N$) — the first and last rounds of the longest winning streak.

If there are multiple longest winning streaks, output the one with the maximum l.

You can obtain partial points if you output any longest winning streak (not necessarily with the maximum l). See the Scoring section for more details.

Scoring

For all tests, the input will satisfy the following constraints:

- $1 \le N \le 5 \times 10^5$
- $0 \le A_i \le 1$ for all $1 \le i \le N$.
- Evirir won at least one round, that is, $A_i = 1$ for some $1 \le i \le N$.

In any subtask, if your answer for every test is a longest winning streak, but for some test it is not the one with the maximum l, you get 50% of the subtask's points.

Output Task

Subtask	Output file(s)	Points	Additional constraints

0	`out_sample_01.txt`
	`out_sample_02.txt`
	`out_sample_03.txt`
		0	Samples.

1	`out_01.txt`	20	All 1s come before all 0s in A, that is, $A_i \ge A_{i+1}$ for all $1 \le i < N$

Code Task

Subtask	Points	Additional constraints
0	0	Samples.
1	35	$N \leq 5000$
2	45	—

Examples

standard input	standard output
14 0 1 1 0 1 1 1 0 0 1 1 1 0 1	10 12
5 1 1 1 0 0	1 3
4 1 1 1 1	1 4

Note

Sample 1

This sample is valid for code subtasks 1 and 2.

Evirir played 14 rounds. It lost rounds 1, 4, 8, 9, and 13, and won rounds 2, 3, 5, 6, 7, 10, 11, 12, and 14.

There are two longest winning streaks, both containing 3 rounds: rounds 5 to 7 and rounds 10 to 12. Since $10 > 5$, the answer is the streak with $l = 10$, so output `10 12`. If you output 5 and 7 instead, you can still get 50% of the subtask's points, as long as every other test in the subtask also gets a longest winning streak (see the Scoring section).

The following are **not** longest winning streaks:

- Rounds 5 to 10: not a winning streak, because Evirir lost rounds 8 and 9.
- Rounds 2 and 3: a winning streak, but it only contains 2 rounds, so it is not the longest.

Sample 2

This sample is valid for output subtask 1 and code subtasks 1 and 2.

Sample 3

This sample is valid for output subtask 1 and code subtasks 1 and 2.

Note that Evirir might win all rounds.